iris-pca-r

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Introduction

Principal Component Analysis (PCA) is a widely used dimensionality reduction technique. It transforms a set of possibly correlated variables into a smaller number of uncorrelated variables called **principal components**. Each principal component is a linear combination of the original features, ordered by how much variance (information) it explains in the data.

PCA is commonly used to: - **Reduce dimensionality** while preserving as much information as possible, - **Visualize high-dimensional data** in 2D or 3D (typically PC1 vs PC2), - **Detect patterns and clusters** (e.g., whether groups separate naturally), - **Reduce noise** and redundancy in features (especially when variables are correlated).

In this analysis, PCA is applied to the classic **Iris dataset**, a standard benchmark dataset in statistics and machine learning. The Iris dataset is ideal for learning PCA because: - It contains multiple numeric measurements (4 features), - The dataset is clean and well-structured (minimal preprocessing needed), - It includes a known class label (`Species`) to help interpret separation in PCA space. # Dataset: Iris

The Iris dataset contains **150 observations** of iris flowers from three species: - *Iris setosa* - *Iris versicolor* - *Iris virginica*

For each flower, four numeric measurements are provided: 1. Sepal Length 2. Sepal Width 3. Petal Length 4. Petal Width

Libraries

```r
install.packages("ggplot2")
```

```
## Installing package into '/cloud/lib/x86_64-pc-linux-gnu-library/4.5'
## (as 'lib' is unspecified)
```

```r
library(ggplot2)
```

Load Data

```r
data(iris)
head(iris)
```

```
##   Sepal.Length Sepal.Width Petal.Length Petal.Width Species
## 1          5.1         3.5          1.4         0.2  setosa
## 2          4.9         3.0          1.4         0.2  setosa
## 3          4.7         3.2          1.3         0.2  setosa
## 4          4.6         3.1          1.5         0.2  setosa
## 5          5.0         3.6          1.4         0.2  setosa
## 6          5.4         3.9          1.7         0.4  setosa
```

Selecting Numeric Features

PCA requires numeric input. Therefore, we exclude the categorical label Species and keep only the first four numeric columns.

```
iris_numeric <- iris[, 1:4] #Let's numerically convert the iris dataset.
head(iris_numeric)
```

```
##   Sepal.Length Sepal.Width Petal.Length Petal.Width
## 1          5.1         3.5          1.4         0.2
## 2          4.9         3.0          1.4         0.2
## 3          4.7         3.2          1.3         0.2
## 4          4.6         3.1          1.5         0.2
## 5          5.0         3.6          1.4         0.2
## 6          5.4         3.9          1.7         0.4
```

Running PCA

We use the prcomp() function, which performs PCA via singular value decomposition. scale.=TRUE standardizes features before computing principal components.

```
pca_result <- prcomp(iris_numeric, scale. = TRUE)
```

PCA Summary and Explained Variance

A key part of PCA interpretation is understanding how much variance each component explains. Typically, the first two components (PC1 and PC2) capture the largest amount of information and are therefore most useful for visualization.

```
summary(pca_result)
```

```
## Importance of components:
##                           PC1    PC2     PC3     PC4
## Standard deviation     1.7084 0.9560 0.38309 0.14393
## Proportion of Variance 0.7296 0.2285 0.03669 0.00518
## Cumulative Proportion  0.7296 0.9581 0.99482 1.00000
```

How to interpret the summary output:

Standard deviation shows the spread captured by each component.

Proportion of Variance indicates the fraction of total variance explained by each component.

Cumulative Proportion tells how much variance is explained by the first k components together.

If PC1 + PC2 explain a large proportion, the 2D plot will be a strong representation of the data structure.

PCA Scores (Coordinates in PC Space)

The PCA result contains scores, which are the coordinates of each sample in the new component space. We extract these scores and reattach the species label for visualization.

```
pca_scores <- as.data.frame(pca_result$x)
pca_scores$Species <- iris$Species
head(pca_scores)
```

```
##         PC1        PC2         PC3          PC4 Species
## 1 -2.257141 -0.4784238  0.12727962  0.024087508  setosa
## 2 -2.074013  0.6718827  0.23382552  0.102662845  setosa
## 3 -2.356335  0.3407664 -0.04405390  0.028282305  setosa
## 4 -2.291707  0.5953999 -0.09098530 -0.065735340  setosa
## 5 -2.381863 -0.6446757 -0.01568565 -0.035802870  setosa
## 6 -2.068701 -1.4842053 -0.02687825  0.006586116  setosa
```

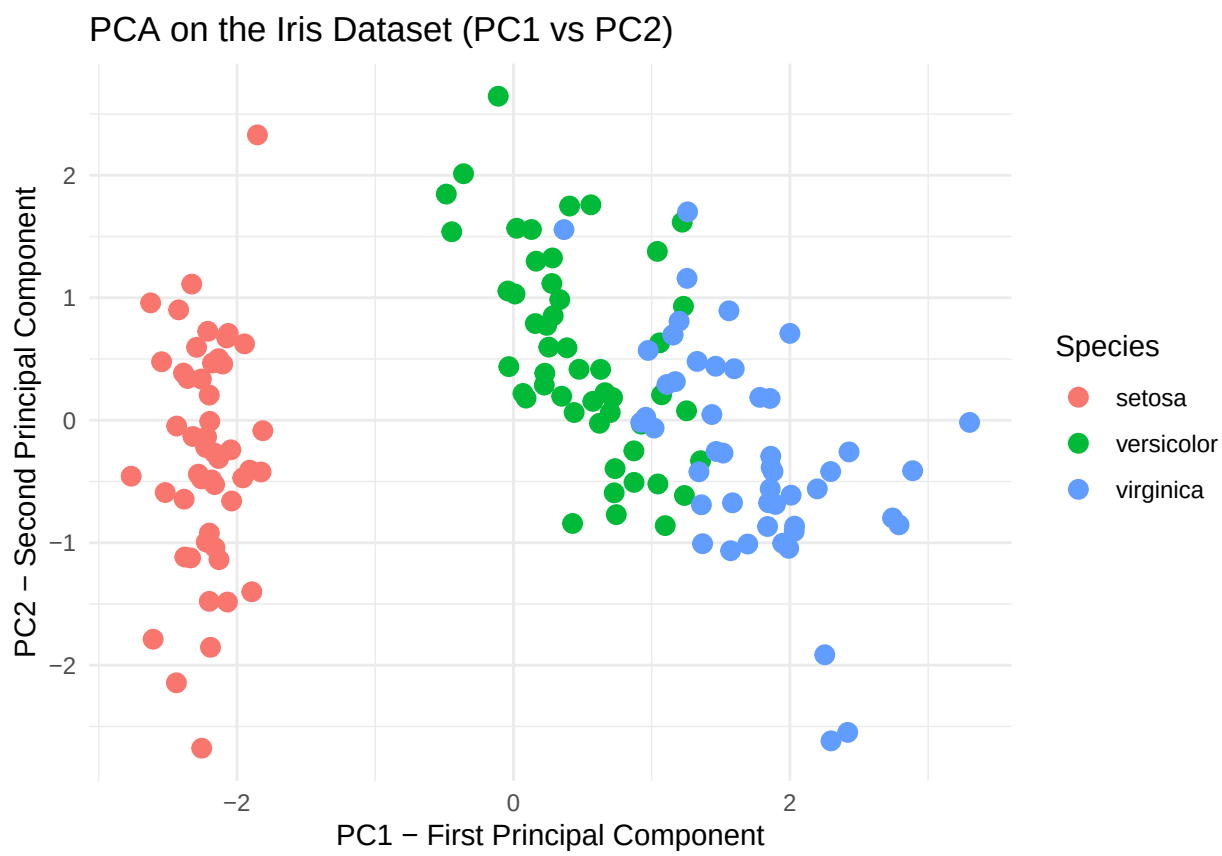
Visualization: PC1 vs PC2

We plot the first two principal components and color points by species.

Interpretation guidelines:

Points closer together represent more similar samples (based on the original features).

Clear separation between colors indicates that PCA captures meaningful structure related to species.

Overlap suggests that those groups share similar measurements in the reduced space.

```
ggplot(pca_scores, aes(x = PC1, y = PC2, color = Species)) +
geom_point(size = 3) +
xlab("PC1 - First Principal Component") +
ylab("PC2 - Second Principal Component") +
ggtitle("PCA on the Iris Dataset (PC1 vs PC2)") +
theme_minimal()
```